Assignment 01 – DSA II Lab

Total Marks: 15 + 10(Viva)

Strict Deadline: May 2, 2026 (11:59 pm)

Strict Academic Integrity Warning

You are strictly prohibited from copying code from classmates, the internet, or any other external sources. You are strictly prohibited from using any generative AI tools to write, refactor, or structure your code. Furthermore, the viva is specifically designed to verify your authorship. If you cannot confidently explain the code you submitted, it will be treated as an academic integrity violation, resulting in an automatic zero for the assignment and further disciplinary action as per university policy.

Question 1: The Closest Pair of Points

Problem Statement: You are given an array of n points on a 2D plane, where each point is represented as a coordinate pair [x, y]. Your task is to find the pair of points that have the smallest Euclidean distance between them and return that minimum distance.

The Euclidean distance between two points $p_1=(x_1, y_1)$ and $p_2=(x_2, y_2)$ is calculated using the standard formula:

$$d(p_1, p_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Requirement: Your solution must utilize a Divide and Conquer approach.

Input:

- A 2D array (or list of lists/tuples) points of size n, where points[i] = [x_i, y_i].
- Coordinates can be negative, positive, or zero.

Output:

- A single floating-point number representing the minimum Euclidean distance between any two distinct points in the input array.

Example Cases

Example 1

- **Input:** points = [[2, 3], [12, 30], [40, 50], [5, 1], [12, 10], [3, 4]]
- **Output:** 1.41421
- **Explanation:** The two closest points are [2, 3] and [3, 4]. The distance between them is $\sqrt{(2-3)^2+(3-4)^2} \approx 1.41421$.

Example 2

- **Input:** points = [[0, 0], [0, 5], [0, 10], [0, 12], [0, 20]]
- **Output:** 2.00000
- **Explanation:** All points lie on the y-axis. The closest pair is [0, 10] and [0, 12], with a distance of 2.

Example 3

- **Input:** points = [[1, 1], [-1, -1], [-1, 1], [1, -1], [0.5, 0.5]]
- **Output:** 0.70710

Question 2: Radar Installation

Problem Statement: Assume a straight coastline acts as the x-axis $(y=0)$ of a 2D Cartesian coordinate system. The open ocean lies in the region where $(y>0)$, and the land lies where $(y \leq 0)$.

There are n small islands located in the ocean, each represented by a 2D coordinate [x, y]. To monitor the ocean, the coast guard wants to install radar systems along the coastline (meaning every radar must be placed perfectly on the x-axis, where $(y=0)$.

Each radar system has a maximum scanning radius of d. An island at coordinate (x, y) is considered covered by a radar located at $(c_x, 0)$ if the Euclidean distance between them is less than or equal to d. Mathematically, this means:

$$\sqrt{(x-c_x)^2+(y-0)^2} \leq d$$

Your task is to write an algorithm to find the minimum number of radar installations required to cover all n islands. If it is mathematically impossible to cover even a single island (because it is located too far out at sea), your algorithm should return -1.

Requirements: Your solution must utilize a Greedy Strategy.

Input:

- An integer d representing the scanning radius of the radars.
- A 2D array islands of size n, where islands[i] = [x, y] represents the coordinates of the *i*-th island.
- All y-coordinates will be positive integers (y > 0).

Output:

- A single integer representing the minimum number of radars needed. Return -1 if it is impossible to cover all islands.

Example Cases:

Example 1: Single Radar Coverage

- **Input:** `d = 5`, `islands = [[0, 3], [4, 3], [8, 3]]`
- **Output:** `1`
- **Explanation:** A single radar placed at coordinate `[4, 0]` can cover all three islands.
  - Distance to `[0, 3]` is $\sqrt{(0-4)^2+(3-0)^2} = \sqrt{16+9} = 5$.
  - Distance to `[4, 3]` is $\sqrt{(4-4)^2+(3-0)^2} = \sqrt{0+9} = 3$.
  - Distance to `[8, 3]` is $\sqrt{(8-4)^2+(3-0)^2} = \sqrt{16+9} = 5$.
    All distances are ≤ 5.

Example 2: Multiple Radars Needed

- **Input:** `d = 5`, `islands = [[0, 3], [10, 3], [20, 3]]`
- **Output:** `3`
- **Explanation:** The islands are spread too far apart.
  - The first island requires a radar somewhere between `x = -4` and `x = 4`.
  - The second island requires a radar somewhere between `x = 6` and `x = 14`.
  - The third island requires a radar somewhere between `x = 16` and `x = 24`.
    Because these required intervals do not overlap, you must install 3 separate radars.

Example 3: Impossible Coverage

- **Input:** `d = 2`, `islands = [[1, 2], [3, 4], [5, 1]]`
- **Output:** `-1`
- **Explanation:** The second island is located at `y = 4`. Because the radar radius is only 2, and the radars must be placed on the x-axis (`y = 0`), the closest a radar could ever get to this island is a distance of 4. Therefore, it can never be covered, and the task is impossible.

Submission Guidelines

Please follow these instructions exactly to ensure your assignment is graded properly. Incorrectly formatted submissions may receive a penalty or be rejected.

1. **Create a Folder:** Place your two completed C++ source files (e.g., `ClosestPair.cpp` and `RadarInstallation.cpp`) into a single new folder.
2. **Rename the Folder:** Rename this folder to exactly match your University Student ID.
3. **Compress the Folder:** Compress the renamed folder into a `.zip` archive.
4. **Submit:** Upload and submit this single `.zip` file to the ELMS portal before the deadline.

Viva Evaluation (10 Marks)

Please note that submitting the code is only the first part of this assignment. There will be a **mandatory viva (oral examination)** based entirely on your submitted code, which will bear **10 marks** of your overall assignment grade.

During the viva, you will be asked to:

- Explain the step-by-step logic of your implementations.
- Walk through the time and space complexity of specific blocks of your code.